Predicting Offense Levels Across NYPD Arrests, 2023-25

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Link to Github Rehub:
https://github.com/ashleybrea/Weighted-Scales-Demographic-Disparities-in-NYPD-Arrests

Introduction & Motivation:

Crime rates in New York City reflect socioeconomic conditions and biases present within the city, as well as crime policies. Arrests made by NYPD officers have received criticism and made headlines citing use of excessive force as well as potential biases towards marginalized groups that impact arrest rates. Data on crime rates and arrests conducted by the NYPD are helpful in determining the work that needs to be done to reduce crimes in NYC, as well as if current policies towards crimes are helpful in solving such a vast problem. Using NYPD Historic Arrest Data (2006-Present), the question we're answering is: How does the relationship between demographic information and level of offense (felony, misdemeanor, violation) vary across NYC's boroughs?

Currently, research has been done on NYPD arrests using predictive analytics to identify patterns and to forecast future crime occurrences. Previous research has found that drug-related offenses committed by people aged 25-44 are the most common in NYC (Kumar et al, 2024). While Kumar et al.'s research focuses on general criminal trends in hotspots and contextualization based on socioeconomic status, we aim to conduct this research with the goal of uncovering biases. Further, while this research uses NYPD data from 2006-2017, we will be contextualizing criminal offenses from 2023-2024, which can provide insight on recent policy changes and social trends. Similarly, racial biases within NYPD arrests have been researched, specifically the use of the Stop-and-Frisk program and creating a bias-free dataset to prevent future biased stops (Badr & Sharma, 2022). Our research project will use similar techniques to both of these research papers, utilizing K-Nearest Neighbors and Random Forest Classification, as well as the construction of neural nets using scikit-learn's Multi-layer Perceptron classifier (MLP).

Our motivation in conducting this research comes from systemic issues of racism within NYPD's criminal system and the disenfranchisement of marginalized groups using the prison system. By conducting this research, we aim to uncover biases within NYPD arrests and contribute our analysis to research that can predict future trends involving crime rates and offense levels, and better understand arrests classification's impact on marginalized groups. Our goal with this project is to analyze the relationship between demographic information (like race and gender) and level of criminal offense. We aim to uncover trends in 2023 and 2024 by conducting this research, and expect that our analysis will assess whether elements like race, borough, and sex are enough demographic information to feed a model that can accurately predict offense level. We also expect to better understand the distribution of our data as it relates to these demographic elements.

Data Description:

Our primary dataset, entitled "NYPD Arrests Data (Historic)" from NYC OpenData, includes all NYPD arrests made to date since 2006. The data was provided by the NYPD itself and contained nearly 6 million entries before cleaning.

Variable descriptions (when sourced from NYC OpenData):

ARREST_DATE	Exact date of arrest for the reported event
LAW_CODE	Law code charges corresponding to the NYS Penal Law, VTL and other various local laws
ARREST_BORO	Borough of arrest. B(Bronx), S(Staten Island), K(Brooklyn), M(Manhattan), Q(Queens)
ARREST_PRECINCT	Precinct where the arrest occurred
AGE_GROUP	Perpetrator's age within a category
PERP_SEX	Perpetrator's sex description
PERP_RACE	Perpetrator's race description

Initial Cleaning:
Given the sheer size of the original dataset, we decided to begin by requesting a singular ten year span (2015 to 2025), so we could potentially explore specific questions related to time periods. We queried our data when extracting it from the website to only include arrests from this ten year span. We removed rows with missing values in any of our above columns, and rows that described the arrests of minors. Our ARREST_DATE was transformed to a datetime data type, and our other variables were cast to a category data type. To help us easily focus on certain parameters, we removed columns we do not foresee using in future analyses, including various classification codes the NYPD included and multiple ways to measure latitude and longitude where a crime took place (although these columns might be of interest to researchers in the future).

Further Data Preparation:
During the modeling process, we realized additional cleaning was required beyond our initial proposal phase. Upon inspecting LAW_CAT_CD further, we found that in addition to the expected classes (F, M, V), the column contained invalid values such as “I”, “9”, and “null”. These were filtered out before training. We also confirmed that LAW_CAT_CD was the correct target variable for our research questions rather than LAW_CODE, which has thousands of varying law codes, which we thought was not appropriate for our classification.

After realizing the categorical nature of of some of features would be incompatible with the models we were attempting to train, we decided to convert those features (PERP_RACE, PERP_SEX, AGE_GROUP, and ARREST_BORO) using the pandas method .get_dummies(), understanding that this comes with potential limitations of creating a sparse dataframe that is slower to train. We originally did the dummy creation manually, but then recalled this method that makes processing more efficient. These variables take values of either 0 or 1, and are labeled in the same fashion: with “IS” followed by an underscore and the associated demographic characteristic. They are used in all models and central to our exploration to see if they are valuable predictors for the offense level of an arrest. For testing across sex, we have one dummy variable “IS_MALE”, so our analysis in this regard is limited to those arrested who were male, and those arrested who were not male.

Because of the sheer size of the dataset and the number of calculations needed to train certain models, like K- Fold Cross Validation, we narrowed the data set before this part of KNN training. Besides slimming down our focus to the years 2023-2024, we used random sampling to narrow the number of data points used in training to 100,000, and tested values of *k* from 1 to 20.

We chose to narrow down our date range because training the model on a decade of data was extremely difficult, especially for the KNN model. While we initially narrowed down our “ARREST_DATE” to 2019 and 2020 for this training phase, we plan to focus on more recent data, taking into account the time it takes to train each model. Thus, our “ARREST_DATE” range is now from 2023-2024.

Methods

We decided to pursue four different model types in our data analysis. We were introduced to Random Forest Classifiers and its associated adaptability through our scholarly research, and knew we could also successfully implement logistic regression, neural nets, and KNN to perform our multiclass classification with discrete numeric variables.

Random Forest Classifier

Model Implementation
We initially trained a baseline Random Forest Classifier (RFC) with default parameters. This resulted in an accuracy of 0.576 but a balanced accuracy of only 0.258. This suggested to us that we should observe the underlying spread of offense levels, and learned that misdemeanors are around 57% of arrests. Violations made up less than 1%. This suggested that standard models would be heavily biased toward predicting the majority class of the data. To address the issue of class imbalance in our RFC, we retrained the model using the “class_weight=’balanced’” parameter. We then attempted some hyperparameter

tuning, such as increasing n_estimators to 200, setting max_depth to 10, and setting min_samples_leaf to 5. This ended up producing no significant improvement.

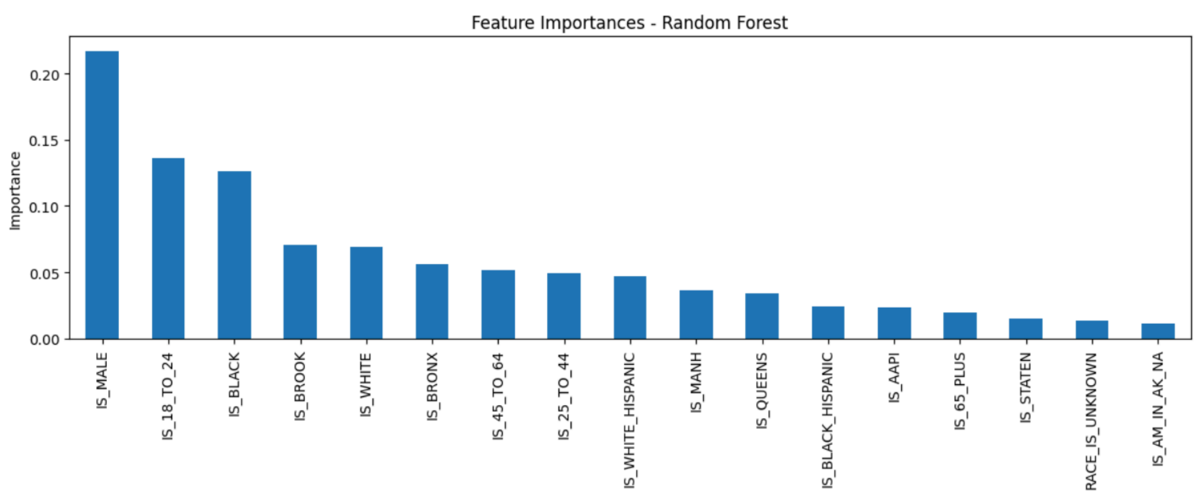
Visualizations/Summary Statistics

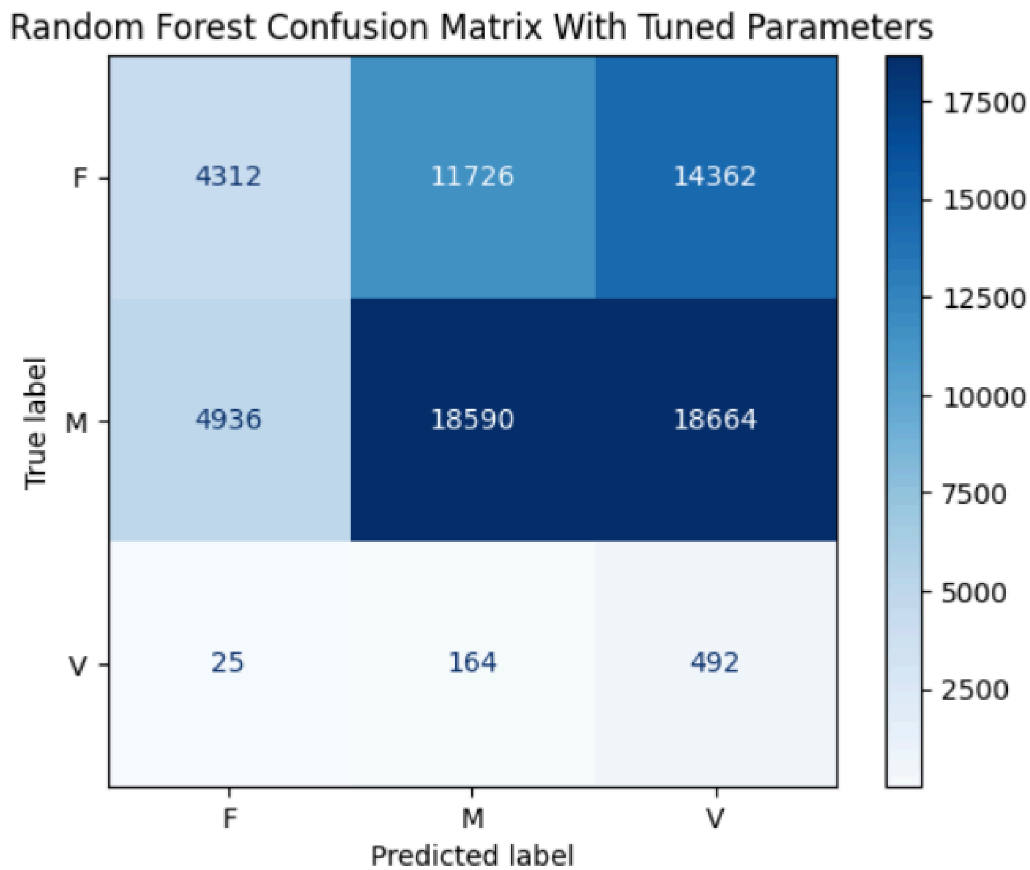
Accuracy before tuning or balancing: `0.5763617633171653`
Balanced accuracy before tuning or balancing: `0.2575581750711388`

Accuracy, balanced: `0.25163111130188437`
Balanced accuracy, balanced: `0.4995534711528792`

Accuracy, tuned and balanced: `0.25359488346138986`
Balanced accuracy, tuned and balanced: `0.4993731820013496`

Below we visualize how important each feature was in our RFC, and our confusion matrix for the tuned RFC model. Notice the discrepancy between volumes of each class.





KNN

Model Implementation

We trained a baseline KNN model using age, race and borough to predict offense level—initally setting our *k* to 1, 3, and 10. We observed that a k of 1 resulted in the best accuracy. We then decided to perform K- Fold Cross Validation using a random selection of data points to reduce calculation time without sacrificing accuracy. This resulted in a best *k* of 13 and a best CV error rate of 0.457. After finding an ideal *k* value of 13, we observed an accuracy score of 0.55 over the test set. While we were not able to illustrate feature performance, we created a confusion matrix to again visualize model performance.

Visualizations/Summary Statistics

Accuracy using ideal *k* value of neighbors (13) : `0.548944070051938`

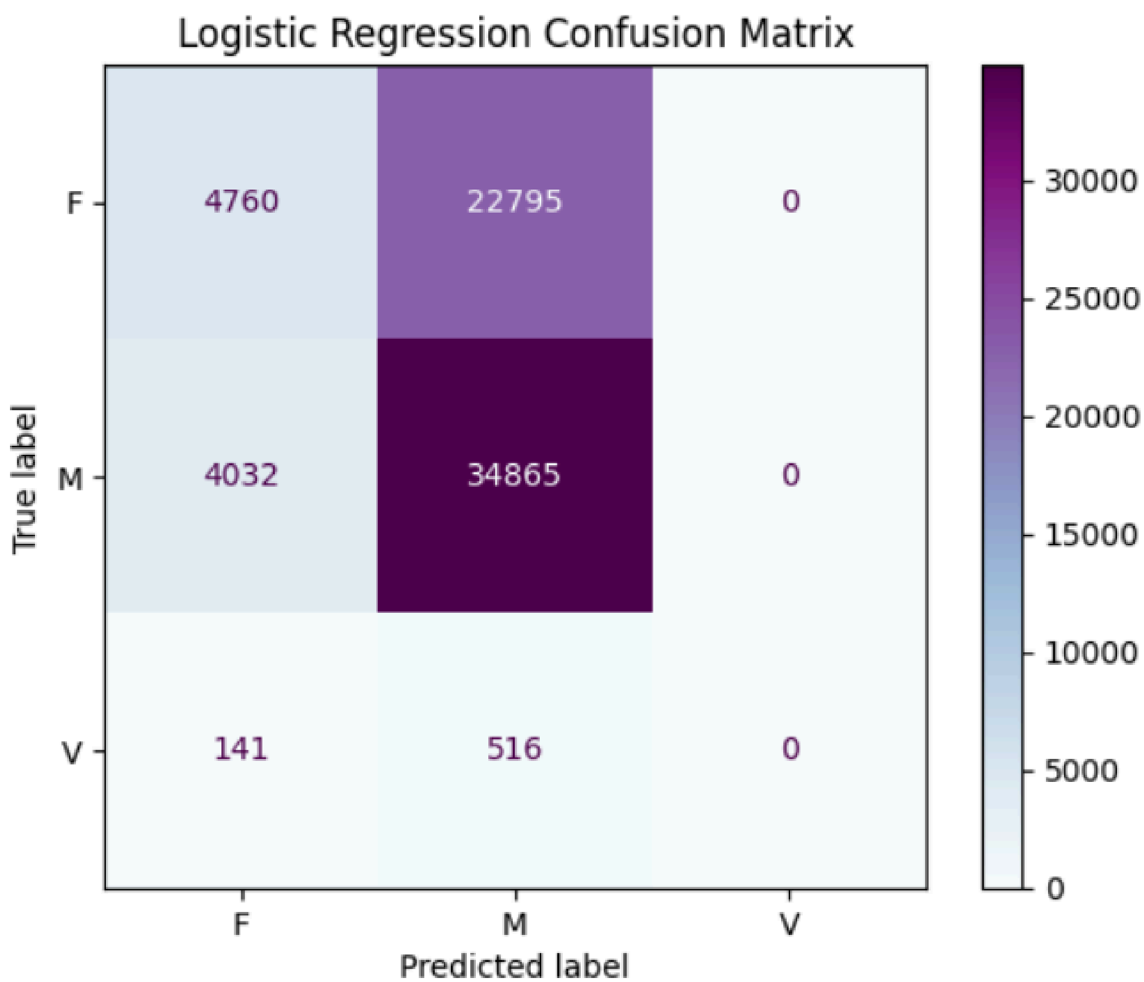
Logistic Regression

Model Implementation

The scikit-learn Logistic Regression algorithm was implemented with max_iter= 5000. The model gained a base accuracy score of 0.574 and a balanced accuracy of 0.254. Features were scaled to account for low accuracy, but the accuracy score didn't change significantly. We then decided to test accuracy rates for the model across race, borough, and sex, which we also executed for the following method (neural networks).

Visualizations/Summary Statistics

Accuracy over entire dataset : `0.5748808859509809`
Balanced accuracy over entire dataset: `0.2541558534688066`



Accuracy Separated By Borough:

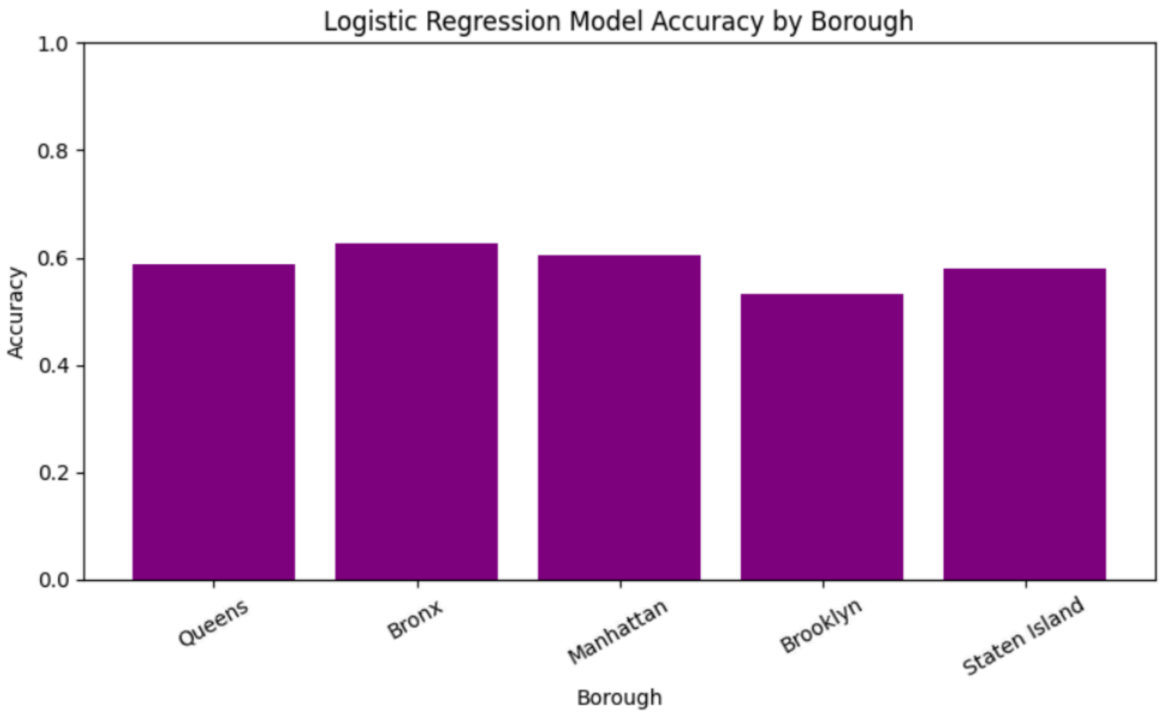
Accuracy in Queens: `0.5868484590257066`
Balanced accuracy in Queens: `0.34468551145478843`

Accuracy in the Bronx: `0.6272933549432739`
Balanced accuracy in the Bronx: `0.3333333333333333`

Accuracy in Manhattan: `0.6055429238616112`
Balanced accuracy in Manhattan: `0.3333333333333333`

Accuracy in Brooklyn: `0.5435883473999455`
Balanced accuracy in Brooklyn: `0.35036661230589866`

Accuracy in Staten Island: `0.5826585695006747`
Balanced accuracy in Staten Island: `0.3333333333333333`



Accuracy Separated By Race:

Accuracy if Black & Hispanic: `0.5772683032334811`
Balanced accuracy if Black & Hispanic: `0.25259662580997544`

Accuracy if Black: `0.5512870292122514`
Balanced accuracy if Black: `0.25666600234208037`

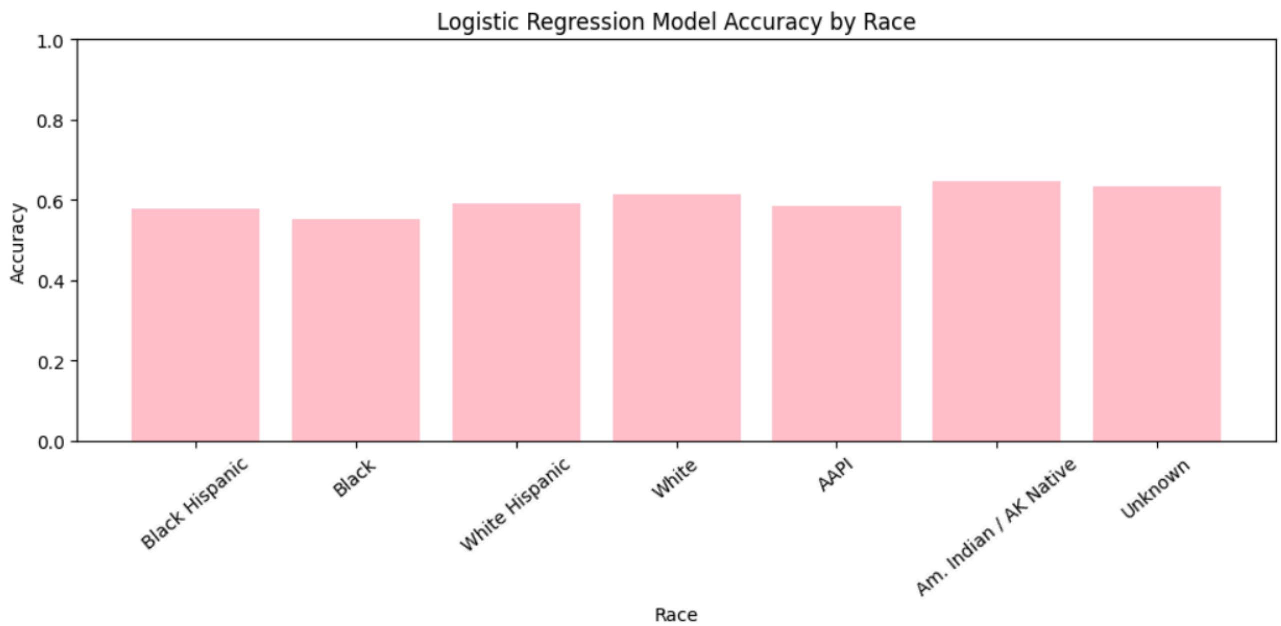
Accuracy if white & Hispanic: `0.5916486563634139`
Balanced accuracy if white & Hispanic: `0.25`

Accuracy if white: `0.6123488224061108`
Balanced accuracy if white: `0.25`

Accuracy if AAPI: `0.5845771144278606`
Balanced accuracy if AAPI: `0.25`

Accuracy if American Indian or Alaskan Native: `0.6472491909385113`
Balanced accuracy if American Indian or Alaskan Native: `0.25`

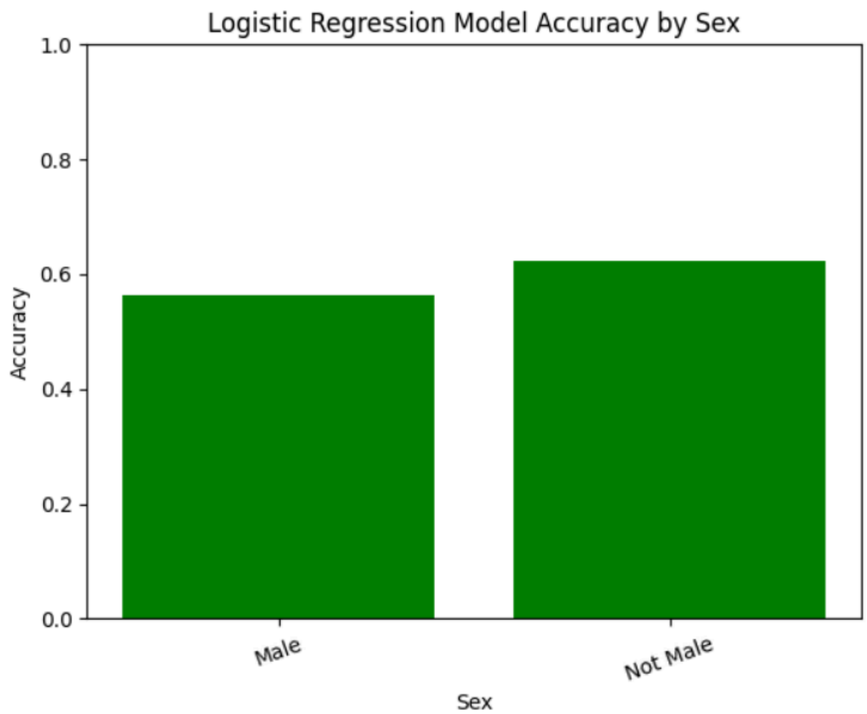
Accuracy if race unknown: `0.6350148367952523`
Balanced accuracy if race unknown: `0.25`



Accuracy Separated By Sex:

Accuracy if male: `0.5623740997722275`
Balanced accuracy if male: `0.25474252153420845`

Accuracy if not male: `0.6222802436901653`
Balanced accuracy if not male: `0.25`



Neural Networks Using MLP

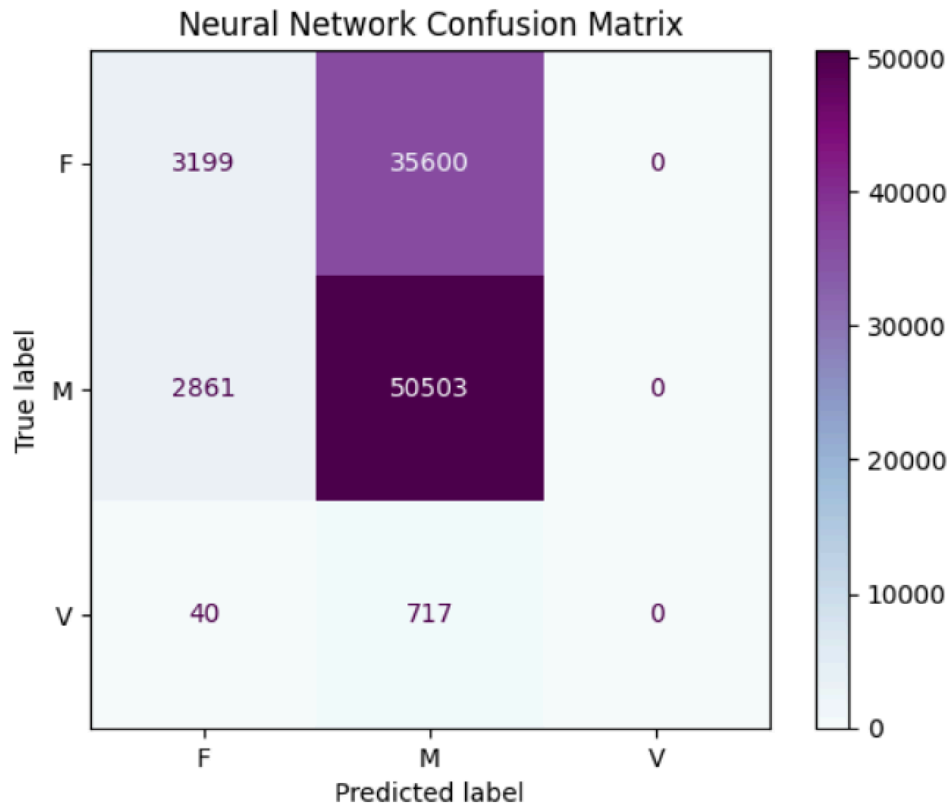
Model Implementation

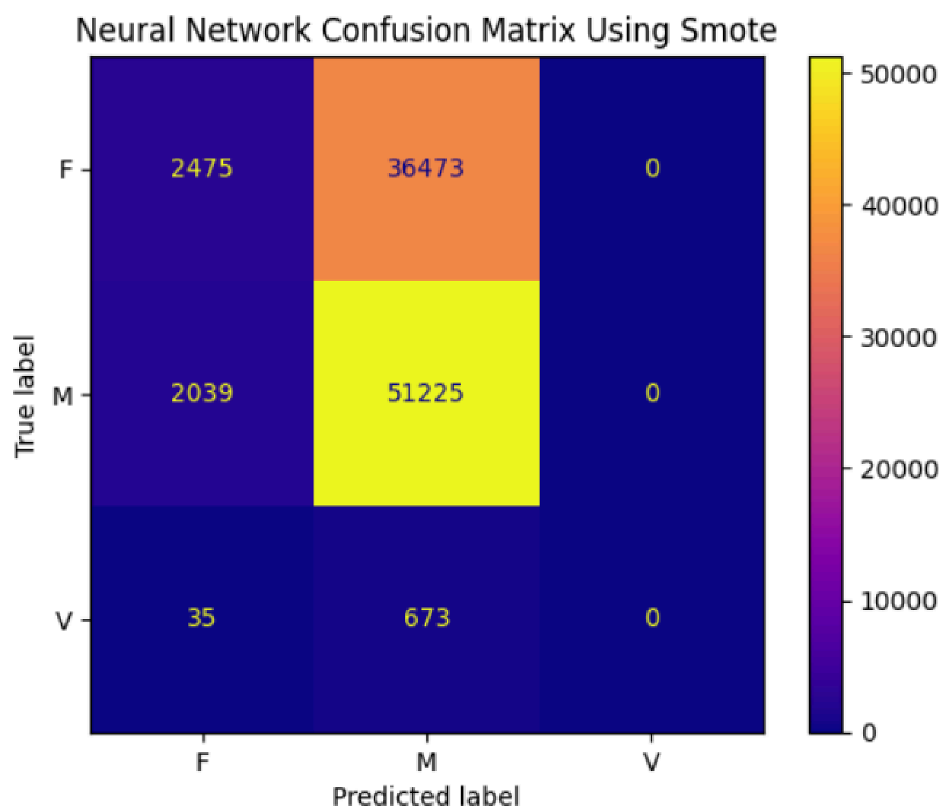
We trained a model using the MLPClassifier method using Adam as an optimization algorithm, with hidden layer sizes of 62 and 34, with max_iter=200. We also produced a confusion matrix, and calculated accuracy by borough, race, and sex. We then attempted using SMOTE (Synthetic Minority Over-sampling Technique) to see if this would remedy any issues with the class imbalance in our data set. The accuracy did not significantly change when training a MLP with a SMOTE dataset.

Visualizations/Summary Statistics

Accuracy over entire dataset : `0.57627591535539082`
Balanced accuracy over entire dataset: `0.25720941401340947`

Using SMOTE, accuracy over entire dataset : `0.576254453363094`
Using SMOTE, balanced accuracy over entire dataset: `0.25631631287434475`


Neural Network Confusion Matrix
True label
F
M
V
3199
35600
0
2861
50503
0
40
717
0
Predicted label
50000
40000
30000
20000
10000
0


Neural Network Confusion Matrix Using Smote
True label
F
M
V
2475
36473
0
2039
51225
0
35
673
0
Predicted label
50000
40000
30000
20000
10000
0

Accuracy Separated By Borough:

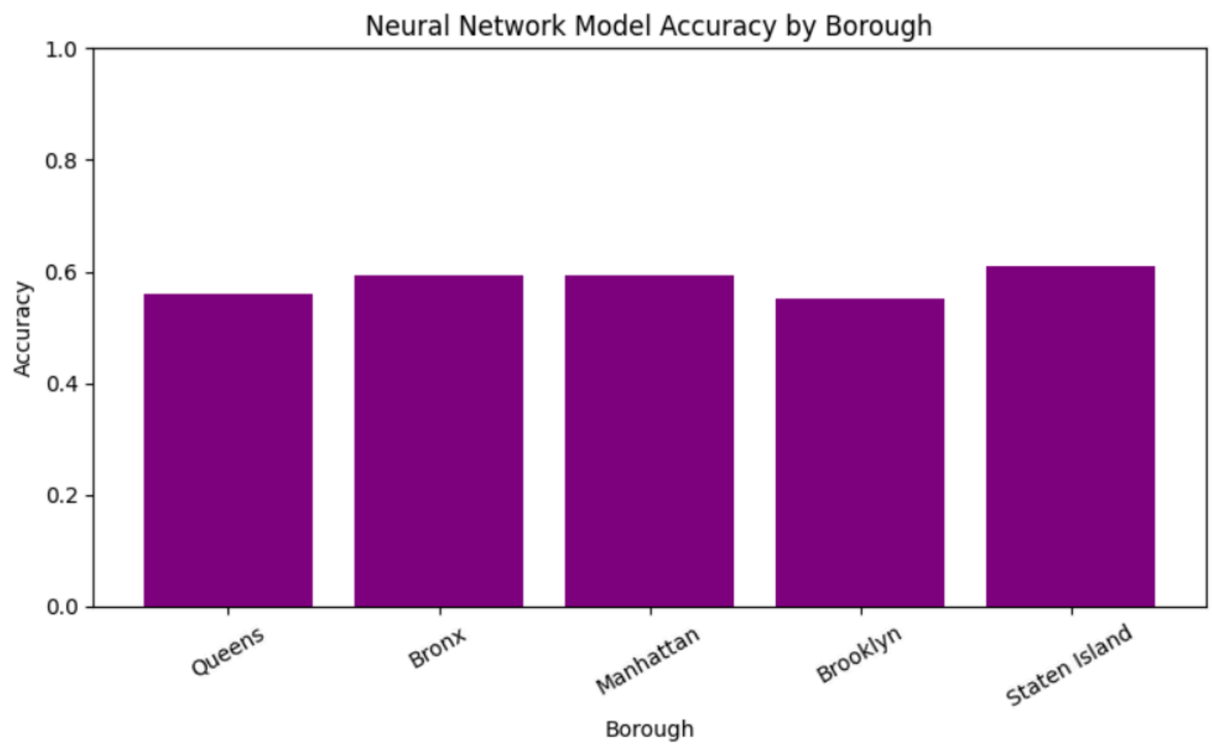
Accuracy in Queens: `0.5611089030206677`
Balanced accuracy in Queens: `0.25676933417960496`

Accuracy in the Bronx: `0.5931733884219916`
Balanced accuracy in the Bronx: `0.2576604201274496`

Accuracy in Manhattan: `0.5929360425473885`
Balanced accuracy in Manhattan: `0.259151340032283`

Accuracy in Brooklyn: `0.5510990079751021`
Balanced accuracy in Brooklyn: `0.2582274569657167`

Accuracy in Staten Island: `0.6112891298989401`
Balanced accuracy in Staten Island: `0.2568434554501064`



Accuracy Separated By Race:

Accuracy if Black & Hispanic: 0.5779171623229156
Balanced accuracy if Black & Hispanic: 0.2527311584099982

Accuracy if Black: 0.5555580933238928
Balanced accuracy if Black: 0.2623811786893279

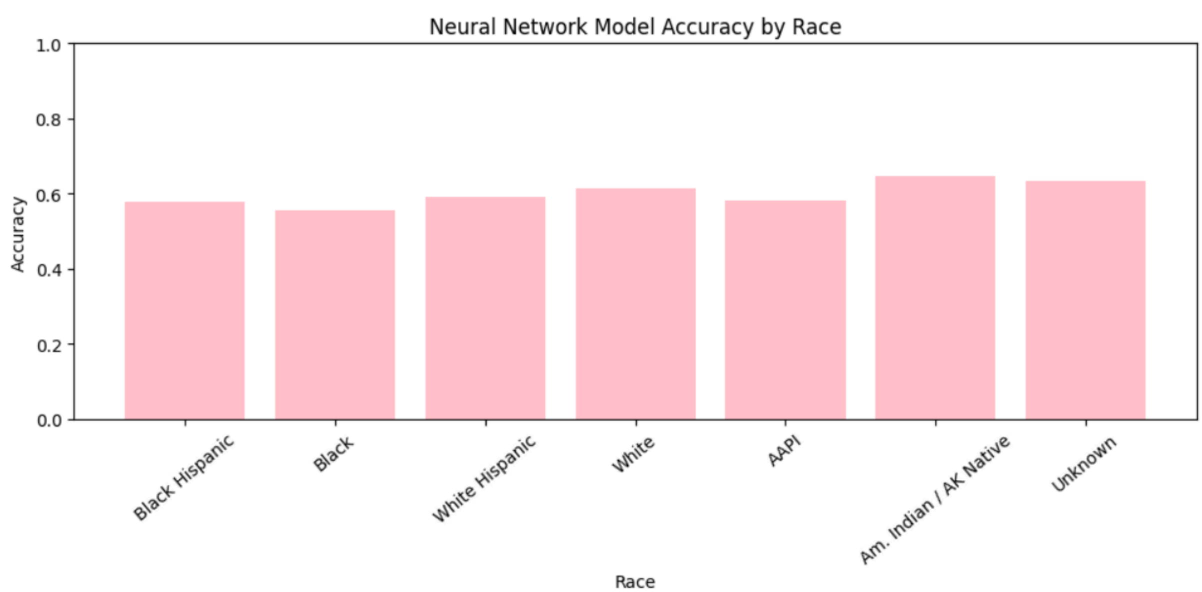
Accuracy if white & Hispanic: 0.5916486563634139
Balanced accuracy if white & Hispanic: 0.25

Accuracy if white: 0.6123488224061108
Balanced accuracy if white: 0.25

Accuracy if AAPI: 0.5818982013011864
Balanced accuracy if AAPI: 0.2547369167275406

Accuracy if American Indian or Alaskan Native: 0.6472491909385113
Balanced accuracy if American Indian or Alaskan Native: 0.25

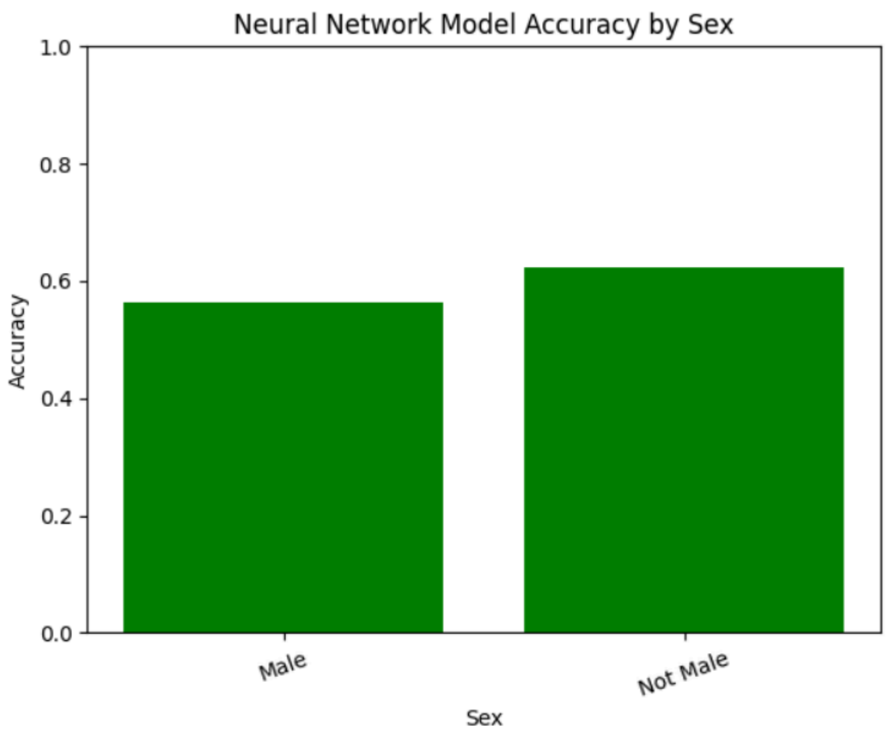
Accuracy if race unknown: 0.6350148367952523
Balanced accuracy if race unknown: 0.25



Accuracy Separated By Sex:

Accuracy if male: `0.5647308203757587`
Balanced accuracy if male: `0.25885737271135806`

Accuracy if not male: `0.6222802436901653`
Balanced accuracy if not male: `0.25`

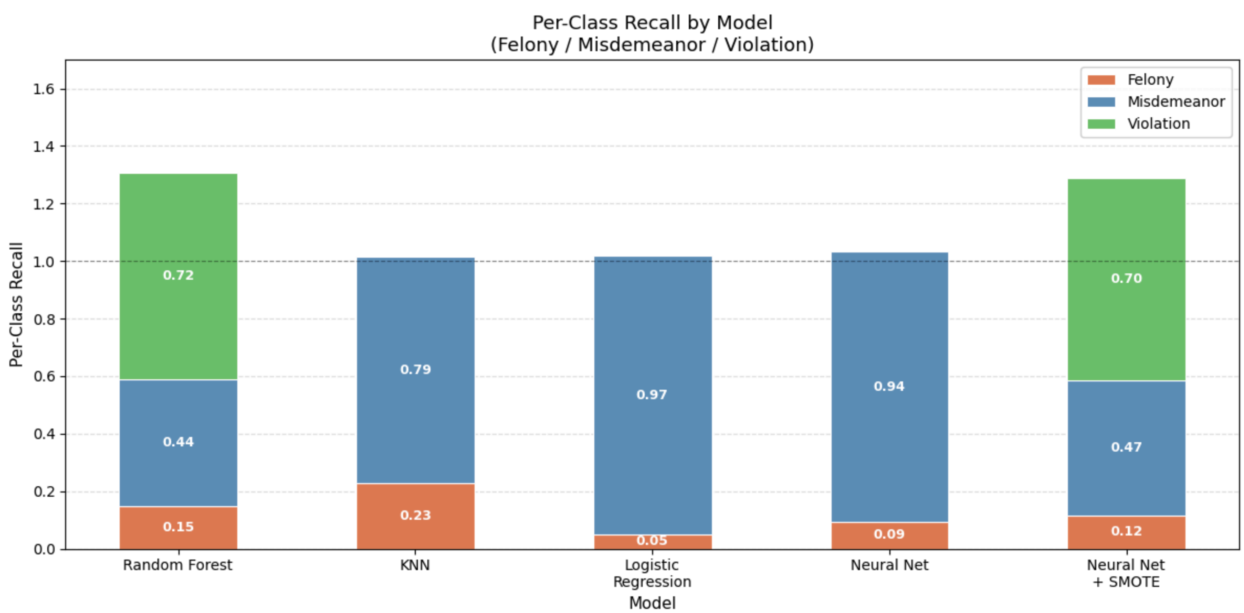


Results

Overall, we saw that overall accuracy in models remained less than 60%, with most models having an accuracy in the range of 55-58%. Balanced accuracy was much lower, ranging between 25%-50%, with Random Forest Classifier having the highest balanced accuracy. This shows that the models are not accurate and are biased towards misdemeanors, which are the majority class of offenses, since the balanced accuracy is so low compared to the overall accuracy. For Random Forest Classifier, the baseline accuracy (57.6%) dropped after applying class balancing (25.2%), while balanced accuracy increased, indicating that class dominance of misdemeanors greatly affected our baseline accuracy. Similar trends followed for the rest of the models. Through analyzing the Logistic Regression model by subgroups of race, borough, and sex, we noticed that overall accuracy was highest in Brooklyn (54%) and Bronx (63%), but balanced accuracy was low between all subgroups, which remained consistent in other models too.

The performance of our models suggest to us that our models perform poorly across all demographics, and our overall accuracy is due to our misdemeanor class majority, which our models classify most of our data as. This is because between all races and subgroups, all of our models perform poorly when it comes to balanced accuracy, even if they receive a high overall accuracy. As a result of our models performing poorly, we conclude that our features don't have enough power to predict level of offense. Further, any variation in level of offense between boroughs is not due to any actual trends from demographic features, but due to data imbalance.

Before we arrived at our results, we tried different methods hoping to answer questions about societal trends in NYC based on policy change and year. We ended up not pursuing this research question due to limited time as well as complexity. We had to narrow down our scope from analyzing arrest records between 2015-2025 to arrest records from 2023-2024 in order to run and train our models. Training our models on a decades' worth of data proved to be too time-consuming, and our computers and Google Colab were not equipped to handle this much data. We decided on narrowing our scope to the two most recent years instead of looking at the pandemic and policy changes between 2015-2025. We also wanted to predict future trends in criminal arrests made by the NYPD across all boroughs, but we were unable to do so because we didn't find any trends between demographic features and level of offense, which was an important precursor to predicting future trends. Along with our research questions, we initially implemented KNN, Random Forest Classification, and Logistic Regression. However, when we calculated accuracies below 60% for all models, we decided to train a Neural Network to see if it could handle data and features that were more complex. We also calculated balanced accuracies for all models after noticing a huge discrepancy between level of offenses in the dataset, with most offenses being misdemeanors and only 1% being violations.



Discussion & Conclusion

Our results connect to a broader conversation about using algorithms in criminal justice. Chouldechova (2017) showed that tools like COMPAS, which predict whether someone will reoffend, face a fundamental problem: when different racial groups have different base rates of reoffending, it is actually impossible for a model to be equally fair to all groups at the same time. COMPAS ends up with false positive rates for Black defendants nearly twice as high as for White defendants, not because the model is poorly built, but because of this unavoidable tradeoff. Our project ran into a similar issue in a different form. Because violations make up less than 1% of our data, no model could reliably predict them no matter how well it was tuned.

One important distinction is that we chose to predict offense level at the time of arrest rather than future criminal behavior. This matters because Chouldechova's work shows how getting that prediction wrong can directly hurt people, thus leading to harsher sentences or denied bail. Our null result is actually somewhat reassuring in that sense by suggesting that demographic information alone isn't driving how serious a charge someone receives.

There are real limits to what we can conclude here. Our features were fairly basic since we didn't include things like the specific type of offense, which precinct made the arrest, or any prior arrest history, all of which would likely improve predictions. It's also worth remembering that this dataset records what officers decided to do, not an unbiased picture of crime, and the offense level shown is the charge at arrest, which can later be changed or dropped entirely. Future work that includes more features or looks at narrower offense categories within felonies and misdemeanors could find patterns we missed, but any time predictive tools are used in criminal justice settings, the kind of careful fairness analysis Chouldechova outlines should come first.